



TYPE CV & SV OIL-IMMERSED ON LOAD TAP CHANGER

Shanghai Huaming Power Equipment Co.,Ltd.



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CV & SV on-load tap changer is a selector switch type of OLTC, that is its change-over selector and diverter switch are composite in the oil compartment and separated from transformer main tank.

CV & SV OLTC is designed and produced complying with IEC-60214-1(2003), it is applicable to the oil immersed transformer for regulating the voltage under the road.

Three-phase CV & SV in Y-connection is only used at neutral end of star connection transformer, 3-phase D-connection CV & SV is for any 3-phase transformer winding connections, and single-phase type CV & SV can be applicable for any selectable transformer winding connection.

Major structure features of the product



- Adopting principle of resistor fast-switching, arc extinguished at first zero point of current.
- Main arc contacts adopt roller structure. Contact life reaches over 100,000 operations when operating at 1.2 times of rated capacity.
- Changing of tapping is smoothly.
- Adopting low pressure die-casting, all aluminum cast parts have good appearance and high mechanical strength.



- Adopting spring energy accumulating mechanism, fast transmission, simple structure.
- Mechanical life of all parts over 800,000 operations.



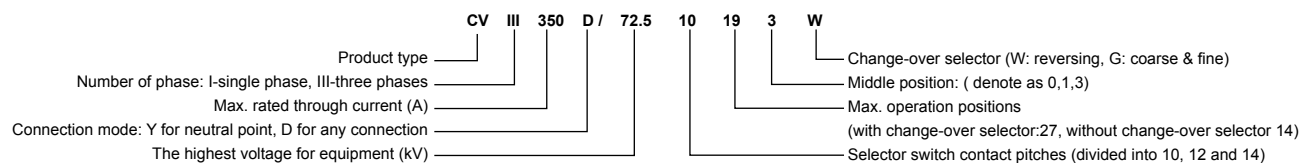
- Excellent oil compartment and oil-resistant seal materials and structure ensures no leakage at 0.03Mpa service pressure.
- Insulation cylinder made of high quality materials, no warp, high dielectric strength.
- Low height, easy connection to transformer.



SV500 shaft developed based on CV500. Its advantages:

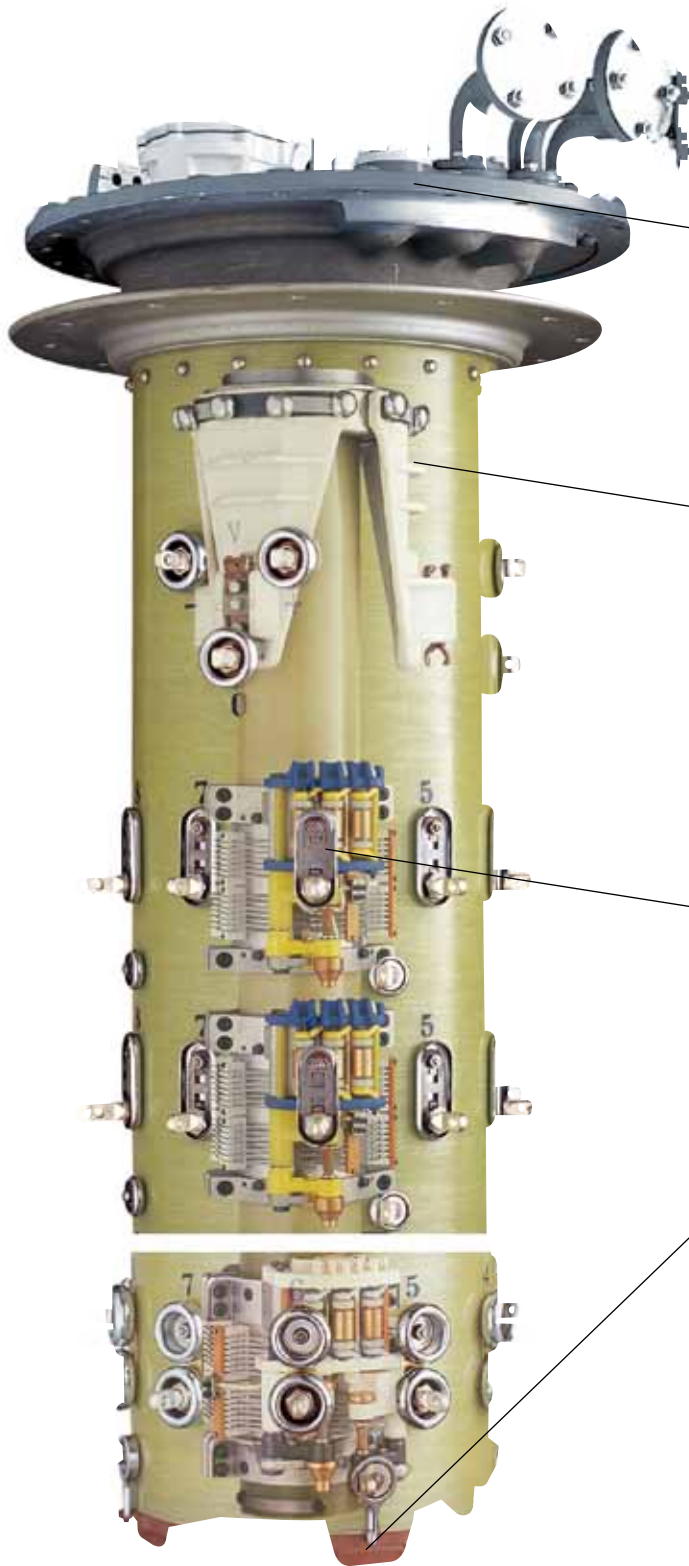
- Structure of main contacts changed from cross to separate, increased moving distance and shortened length of main shaft.
- Adopting clamp structure, support and shaft combined closely and reliably.

Explanation of Type



Major components

Head of tap changer, Oil compartment, Main shaft, Bottom of cylinder



Head of tap changer

All aluminum casting parts adopt low pressure die-casting with good appearance. Grain structure is dense. Mechanical strength is high, no leakage.

Oil compartment

Oil compartment is made of epoxy glass fiber, wound insulation cylinder. It has good insulation and mechanical properties. Oil compartment keeps switch separated from transformer oil. It is also support of stationary contacts.

Main shaft

Two kinds of contacts are mounted on the wall of cylinder. Upper is change-over selector, below is three same groups of moving contacts of tap changer.

Bottom of cylinder

The bottom of cylinder pressed by glass fiber reinforced unsaturated polyester, it has high strength and is easy to be mounted or dismantled.



CV & SV Series OLTC Main Technical Specifications

Model	OLTC models		CVIII350Y		CVIII350D		CVI350		SVIII500Y		SVIII500D		CVI700	
1	Max rated through-current (A)		350				500				700			
2	Rated frequency (Hz)		50 or 60											
3	Number of phase		3		3		1		3		3		1	
4	Connection application		Y -in the neutral, D and single pole - in any part of the winding											
5	Short-circuit current test (kA)	Thermic (3 secs)	5				7				10			
		Dynamic (peak value)	12.5				17.5				25			
6	Max. step voltage (V)	10 contacts	1500				1500				1500			
		12 contacts	1400				1400				1400			
		14 contacts	1000				-				1000			
7	Rated switching capacity (kVA)	10 contacts	525				400 ~ 525*				660			
		12 contacts	420				325 ~ 420*				520			
		14 contacts	350				-				450			
8	Max. operating positions	Without change-over selector	14				12				14			
		With change-over selector	27				23				27			
9	Insulation to ground (kV)	Highest voltage for equipment Um	40.5				72.5							
		Rated separate source AC withstand voltage (kV/50Hz,1min)	85				140							
		Rated lightning impulse withstand voltage (kV/1.2/50μs)	225				350							
10	Rated withstand voltages of the internal insulation		Refer to section 4.6 of CV & SV Technical Data											
11	Mechanical life		not less than 800,000 operations											
12	Electrical life		not less than 200,000 operations											
13	Oil compartment	Operating pressure	0.03MPa											
		Test pressure	0.08MPa, without any leakage for 24 hours											
		Over pressure protection	bursting cap bursts at 300kPa ± 20% overpressure											
		Protective relay	Setting oil flow speed 1.0m/s ± 10%											
14	Equipped with motor drive unit models		CMA7,CMA9 or SHM											
15	OLTC models		CVIII350Y		CVIII350D		CV I 350		SVIII500Y		SVIII500D		CV I 700	
	Weight (approx. kg)		140		150		120		190		200		130	
16	Oil displacement volume (approx. dm³)	Without change-over selector	135		185		85		205		240		120	
		With change-over selector	165		220		115		235		275		150	
17	Volume of oil filling Vs and oil conservator DV (approx.dm³)		Vs	DV	Vs	DV	Vs	DV	Vs	DV	Vs	DV	Vs	DV
		Without change-over selector	105	14	165	21	60	10	160	20	200	21	85	12
		With change-over selector	130	17	180	22	85	12	185	22	225	26	108	15

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